

HIGHLY RELIABLE DRIVING WORKLOAD ANALYSIS USING DRIVER ELECTROENCEPHALOGRAM (EEG) ACTIVITIES DURING DRIVING

J. Y. KIM*, C. H. JEONG, M. J. JUNG, J. H. PARK and D. H. JUNG

Korea Automotive Technology Institute, 74 Yongjeong-ri, Pungse-myeon, Cheonan-si, Chungnam 330-912, Korea

(Received 19 September 2012; Revised 22 March 2013; Accepted 1 April 2013)

ABSTRACT—Traffic accidents are caused by various factors, which can be classified into human factors, vehicle factors and environmental factors. Recently, human factors have been drawing particular attention as efforts are being made to enhance the safety performance of vehicles and improve road conditions. Driving distraction caused by an increased driving workload is a representative human factor. Various studies in the past have attempted to quantify the driving workload by using EEG activities. However, they have failed to consider vibration properties generated from vehicle engines. A number of noise signals were included in brainwave signal processing, which resulted in a failure to obtain reliable outcomes. Thus, this study suggests driver EEG activities free of vehicle engine secondary vibration in order to develop a method that analyzes the driving workload with high statistical reliability. By using the analytical method developed in this study, standard values of driving workload for straight and left-turn driving that has statistical significance could be calculated. The analytical method for driving workload created by this study can be applied to HVI and road design.

KEY WORDS : Driving workload, Electroencephalogram (EEG), Engine vibration, HVI (Human Vehicle Interface)

1. INTRODUCTION

Traffic accidents generally occur due to human factors, vehicle factors and environmental factors, or complex interactions among these factors. As efforts are being made to enhance the safety performance of vehicles and improve the road conditions to reduce traffic accidents, more attention is being paid to human factors. In fact, the National Highway Traffic Safety Administration (NHTSA) and the Department of Transport (DOT) in the US have reported that 90% of traffic accidents is attributable to human factors and only 3% is caused by vehicle factors (Naim *et al.*, 1995).

A representative human factor would be driving distractions resulting from an increased driving workload. Driving distraction refers to a situation in which a driver's recognition of information needed to perform the driving task safely is delayed due to an event, behavior, people or reduced attention to the driving task (Kim *et al.*, 2010).

According to ROTA's 2005 report, "Road traffic accidents in Korea," 60% of road traffic accidents happening in Korea is associated with driving distractions (ROTA, 2005). Meanwhile, according to the 1998 statistics published by Japan's National Policy Agency, 89% of vehicle collision accidents is related to mistakes on the part of drivers, such as driving distractions (NPA, 1998; Kim and Son, 2011). In order to prevent road accidents, research is needed to measure and reduce the driving workload that causes

driving distractions.

Existing studies on driving workload involve measuring drowsy-driving, driver tension, and driver's abnormal reactions through changes in brainwaves, such as α , β and θ waves that indicate a person's state, and analysis of optimal driving conditions (Kim *et al.*, 2004). Meanwhile, some studies have attempted to examine the driving workload by using a diagnostic method that analyzes various pattern changes (Kim *et al.*, 2001; Kim *et al.*, 2002). However, previous studies have failed to consider the vibration properties generated from vehicle engines, and a number of noise signals were included in brainwave signal processing, which resulted in a failure to obtain reliable outcomes in a deterministic method using average values. In addition, the diagnostic method expresses driving workload with four patterns, which has limitations in being described quantitatively.

As such, this study proposes driver EEG activities free of secondary vibration properties from vehicle engines, and develops a method for driving workload analysis with high statistical reliability. The analytical method for driving workload proposed in this study can be utilized in HVI and road design.

2. SYSTEM TO MEASURE DRIVING AND ROAD CONDITIONS

2.1. Data Acquisition System Settings

Several sensors, cameras and a DAQ module were set up in

*Corresponding author. e-mail: kimjy@katech.re.kr



Figure 1. Set-up for data collection device for road and driving conditions.

a diesel SUV, as shown in Figure 1 below, to collect data for driving and road conditions. A diesel SUV by K Company equipped with electronic stability control (ESC) was used as the test vehicle and its vehicle information, such as its steering angle, vehicle speed, engine speed, and fuel amount, was collected, and several devices were installed inside the vehicle.

A DAQ module, inverter and a power module were installed in the trunk so as not to take up the vehicle's interior space, and a GPS was installed on top of the dashboard to enhance the satellite transmission ratio. A gyro sensor was installed inside the console close to the vehicle's center of mass to ensure data accuracy. For brake pressure sensors, a jig was installed in the middle of the brake line of the four wheels and a sensors were placed on the jig to measure the pressure applied to the wheels directly. The front view camera was installed close to the windshield. In addition, a laser scanner was placed by puncturing a hole in the front grill – this is to measure the distance from the vehicle in front and relative speed. A dedicated lane detection device was also installed to collect information of distance from the tire to the left and right lanes.

Sensors were attached to the driver's body, as shown in Figure 2, in order to measure physiological signals. The frontal lobe EEG sensor, which is responsible for recognition in accordance with the international 10-20 system, was

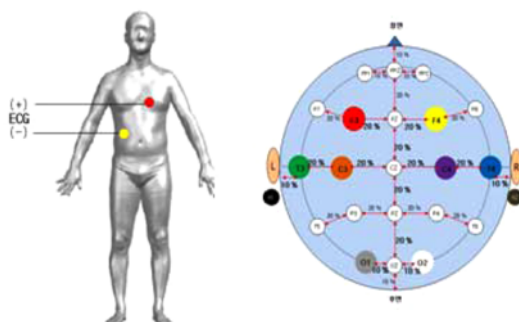


Figure 2. Measuring of ECG signal (left side, standard limb lead method) and EEG signal (right side, international 10-20 system method).

Table 1. Specifications of road and driving conditions from the data collection device.

Name	Model	Specification	Maker
Test vehicle	Sorento R 2.2	TLX	KIA
DAQ module	DEWE-501	8ch analog 8ch bridge 2ch CAN	Dewetron
GPS	NL-302U	reacquisition : 0.1s	navilock
Gyro	MTI	Rate : 300 deg/s	XSENS
Camera	HDCAM-1600UVC	2megapixel 30fps	KRIZER
Inverter	HT-S-1200-12	1200W, DC12V	Izzy power
Brake pressure	PGM200 KE	20 MPa	KYOWA
Lane detection	C2-170 ELD	+/- 635 cm	Mobileye
Laserscanner	ibeoLUX	0.3~200 m	ibeo
Bio measure	Poly G-I	16ch Analog out	LAXTHA
Monitoring S/W	DEWE Soft	Ver7.0.2	Dewetron

placed at F3 and F4 to reduce eye-blinking noise, whereas the EEG sensor for the parietal lobe in charge of movement was placed at C3 and C4. Meanwhile, the EEG sensor for the temporal lobe in charge of hearing was attached to T3 and T4 near the ears, and finally, the EEG sensor for the occipital lobe in charge of vision was placed at O1 and O2. As for the ECG sensor, the positive pole was attached to the heart area and the negative pole was attached to the lower rib diagonal from the heart, and their positions were adjusted in accordance with the potential difference (Koo *et al.*, 2009). A 19-inch monitor, which can be operated with a wireless keyboard, was placed behind the driver's seat for the purposes of test data setting and monitoring. All equipment selected and installed have sufficiently reliable specifications in a real vehicle environment as those shown in Table 1.

2.2. Data Monitoring

As shown in Figure 3, a GUI was created to monitor and store the collected vehicle measurement data. Video images were placed in the front, back, left and right with the driver in the middle for easy recognition, and the engine and vehicle speeds were shown in a similar way as a dashboard, which is a familiar screen for readers. Successive data, such as the steering angle, roll angle, pitch angle, tri-axial acceleration, sound, facial movement and physiological signals were shown in graphs in order to easily recognize overall tendencies. On/off signals, like turn indicators, ABS and ESC activation were visualized by applying circular lamps. Meanwhile, the distance from the vehicle in front, brake pressure and shift position are represented by bar graphs for quick checking of signal changes. GPS data were saved as values of longitude and latitude, and were mapped to Google Earth for image capturing. The video data were saved at the speed of 30fps



Figure 3. Data monitoring system structure for road and driving conditions.

(frames per second) and the GPS data were saved as 1 s/s (sampling per second), whereas the biological signals and other data were saved at 250 s/s and 100 s/s, respectively.

All data were synchronized so that the data could be saved at the same time. When data were measured for about an hour, the volume of data saved was about 1.5Gbytes.

3. DRIVING TEST ON EACH ROAD

3.1. Driving Section

Vehicle driving data for different roads in Korea were collected by using the test vehicle to collect the data of driving/road conditions, which was built for this study. Highways, regional roads and local roads were selected for road types, and the test vehicle ran on each of the roads for about one hour, as shown in Table 2.

For the highways, a section with relatively smooth traffic was chosen; the vehicle was to start from South Cheonan IC on Gyeongbu Expressway, took the Pyeongtaek-Eumseong Expressway at Anseong Junction and then took Jungbu Expressway at Daeso Junction to

Table 2. Driving courses to collect data on road/driving conditions.

Type	Section	Distance (km)	Duration (h)
Highways	South Cheonan IC-Anseong Junction-Daeso Junction-Jincheon IC	76	1.0
Regional roads	Jincheon IC-Baegdok Rest Area-Jangsan Intersection-Cheonan Terminal	50	1.2
Local roads	Cheonan Terminal-Cheonan Station-Lotte Mart-Cheonan City Hall-E Mart-Home Plus	15	0.8
	Total	141	3.0

Table 3. Order of data collection for road/driving conditions.

Duration	Item
13:00~13:20 (20 min)	Briefing and survey
13:20~13:50 (30 min)	Test-driving at KATECH PG
13:50~14:10 (20 min)	Test-driving at South Cheonan IC
14:10~14:20 (10 min)	Break
14:20~15:30 (70 min)	Highway (South Cheonan IC-Jincheon IC)
15:30~15:40 (10 min)	Break
15:40~16:40 (60 min)	Regional road (Jincheon IC-Cheonan Terminal)
16:40~16:50 (10 min)	Break
16:50~17:40 (50 min)	Local road (Cheonan Terminal-Home Plus)
17:40~18:00 (20 min)	Wrap-up and survey

arrive at Jincheon IC. For the regional roads, the course started from Jincheon IC and ended at Cheonan Terminal, which included steep grades and curves. Meanwhile, the local roads' section started from Cheonan Terminal and ended at Home Plus in Cheonan, which was set to run through major landmarks with heavy traffic.

3.2. Test Schedule

Each driver was cautioned for driving and briefing on the three driving sections for 20 minutes starting at 1pm, and afterwards they filled out a survey sheet. Then, the drivers test-drove the vehicle for 50 minutes. Test-driving, albeit short, was necessary to ensure that the drivers were familiar with the test vehicle. Then data were collected from the entrance of the highway, classified by road type, and then collected continuously after a 10-minute break. After the test, the drivers filled out another survey sheet about any discomfort that they may have experienced while driving and items for improvement. The entire test was to last five hours, but the actual duration could vary depending on each driver's driving habits.

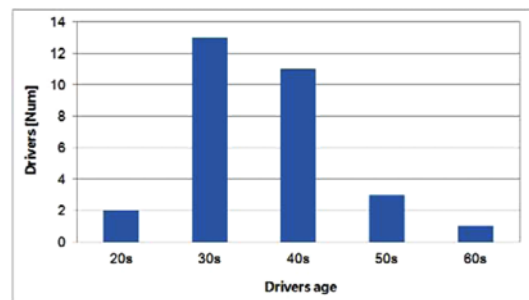


Figure 4. Distribution of the number of drivers by age group.

3.3. Drivers

For this test, 30 ordinary people ranging from their 20s to 60s with over one year experience of driving were selected through open recruitment. The drivers' age distribution was comprised to have normal distribution, as shown in Figure 4, by using the 2009 Police Agency's statistics on traffic accidents as reference.

4. METHOD TO ANALYZE DRIVING WORKLOAD

4.1. Engine Vibration Noise Included in Brainwaves

α waves tend to increase when a driver is relaxed or drowsy, whereas β waves increase when the driver is feeling tensed and uneasy. Thus, a driver's brainwaves were analyzed based on the increase and decrease rate of the relative spectrum of α waves (8~12 Hz) and β waves (13~30 Hz) by using EEG activities (β/α) (Andreassi *et al.*, 1995). The EEG activities, which allow for numerical presentation of tension and uneasiness, can be utilized as quantitative data to determine driving workload (Park *et al.*, 2010).

The frequency range of the β waves overlaps with the secondary vibration properties of the vehicle engine. The secondary properties of the engine are vibration generated from the engine's explosion stroke, and the four-cylindrical engine generates vibration two times in one rotation. When the typical idle speed of 700 rpm is applied, it becomes 23.3 Hz ($700/60 \times 2 = 23.3$ Hz). The secondary vibration properties of the engine are the biggest excitation source in a stopped vehicle, and acts as a noise element to the cables and equipment to measure brainwaves. When the engine starts moving, the secondary vibration properties move too, and this may affect brainwave signals, even while driving. Figure 5 shows the EEG measured when the vehicle was stopped and then processed with fast Fourier transform (FFT). It shows that the secondary vibration properties at 26 Hz when the engine's idle speed is 790 rpm affect the range of β waves.

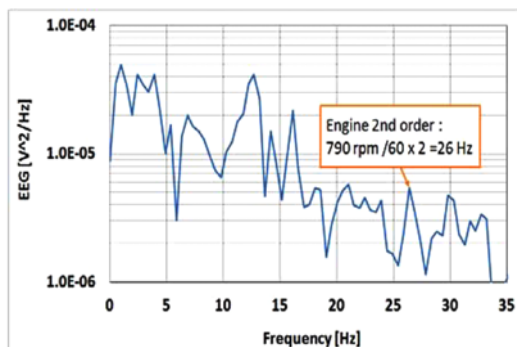


Figure 5. Secondary vibration noise properties of the engine included in brainwave signals (26 Hz at 790 rpm of engine rotation).

Therefore, in this study, the 24-30 Hz section within the range of β waves affected by the engine secondary vibration properties was removed and Half β was defined instead in order to analyze the driving workload. Half β waves had a range of 13-23 Hz, from which secondary vibration values over 700 rpm were removed from a four-cylindrical engine. In order to quantify the driving workload, a driver's brain activities were defined as Equation (1) below by using Half β waves.

$$\text{Driver EEG Activities} = \text{Half } \beta / \alpha \quad (1)$$

4.2. Process to Quantify Driving Workload Using Brainwaves

In order to quantify the driving workload by using brainwave signals, a process to handle the signals, as shown in Figure 6, was established. The sampling of the brainwave signals was set at 250 Hz, and data were collected from DAQ equipment. When sampling was set to 250 Hz, a frequency analysis of up to 125 Hz was possible. The collected brainwave signals were classified by road type. Road types come in pairs – straight section with a left turn, straight section with a right turn, and straight section with a curve. For the classified brainwave signals, 250 blocks were established and overlapped by 50%. Then, frequency analysis was performed by calculating the power spectrum using Hanning Window. For 250 blocks, frequency analysis can be carried out once every second by dividing into sampling periods. By using 50% overlap and Hanning, the frequency analysis noise between data blocks can be reduced. By using the average power spectrum size of 8-

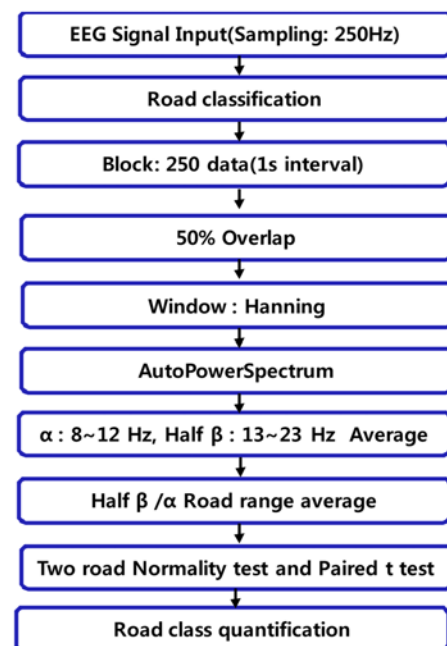


Figure 6. Process to quantify driving workload using brainwaves.



Figure 7. Road classification (straight section with a left turn).

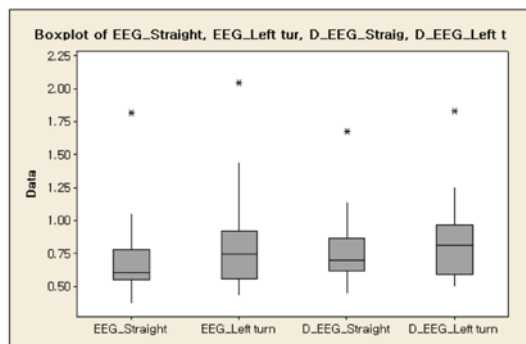


Figure 8. General EEG activities (Left side) and Driver EEG (Right side, D_EEG) activities data boxplot : confirmation of unusual values.

12 Hz, which is the α value, and 13-23 Hz, which is the value of Half β from the frequency analysis results, a driver's EEG activities could be calculated, and the statistical significance of the average based on road type was validated through a normality test and paired t-test.

After obtaining reliability through significance validation, the driving workload for different road types was quantitatively analyzed by using average values.

5. ANALYSIS OF DRIVING WORKLOAD WHILE DRIVING ON A STRAIGHT SECTION WITH A LEFT TURN

Among the driving sections introduced in Chapter 3, the driving workload on a local road with a left turn, as shown in Figure 7, was analyzed by using the method suggested in Chapter 4. The left turn section was divided into two road types – one 800 m-long straight section and another 800 m-long section where the left turn ends - by using the point where the navigation system starts to announce an upcoming left turn or 700 mm from the left turn as the dividing point. A driver's EEG activities were calculated by using the method suggested in Chapter 4, and was shown in a boxplot as demonstrated in Figure 8. One driver exhibited unusual data in terms of general EEG activities

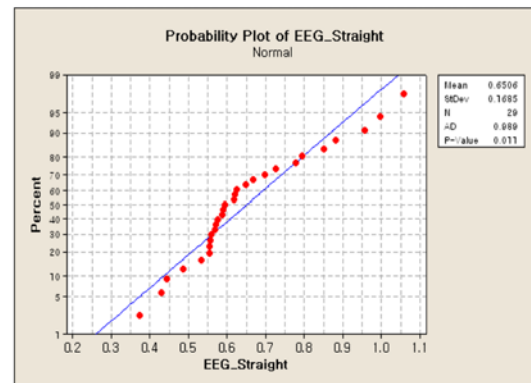


Figure 9. General EEG activities (straight driving) normality test: $P = 0.011$, not normal distribution.

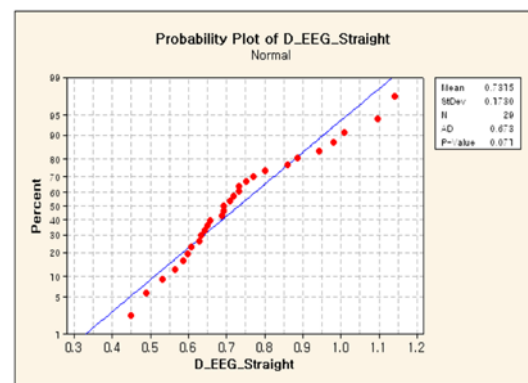


Figure 10. Driver EEG activities (straight driving) normality test : $P = 0.071$, normal distribution.

and driver's EEG activities.

The normality test, as shown in Figure 9 and Figure 10, was performed by using the data of 29 drivers, except the unusual data of one driver. In the EEG activities data for the straight driving by a normal method, the P value is 0.011 and cannot be regarded as normal distribution, whereas in the EEG activities data of drivers in the straight-line mode using the method suggested in this study, the P value is 0.071, which is confirmed as normal distribution. This result indicates that the engine's secondary vibration properties not only affected the range of β waves as noise, but also undermined statistical reliability. By using the drivers' EEG activities that were confirmed to have normal distribution, a paired t-test was performed on the hypothesis that straight driving has a greater driving workload than left-turn driving, as shown in Figure 11. The result was that the P value was 0.008, which represents significance, and the left turn, as shown in Table 4, was analyzed to have a 12% increase in the driving workload (EEG activities) compared to the average value of the straight driving section.

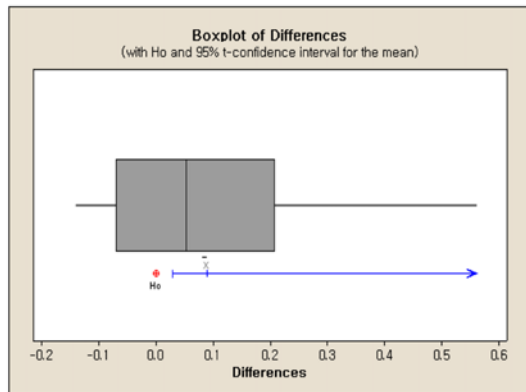


Figure 11. Paired t-test boxplot (Straight and left turn driving).

Table 4. Results of statistical analysis using drivers' EEG activities (left-turn driving has 12% greater EEG activities than straight driving).

Road	Number	Mean	StDev	P Value
Straight	29	0.7315	0.1730	0.008
Left turn	29	0.8202	0.2223	

6. CONCLUSION

This paper suggests the quantitative analysis of brainwaves to examine driving workload, which is one of the human factors that are drawing attention nowadays as a main cause of traffic accidents. It was confirmed that the engine's secondary vibration properties were included in the range of β waves as noise, and the results of the study suggests that the Half β range removed the noise. Also, the driver's EEG activities were suggested by using the ranges of Half β and α waves, and the drivers' EEG data for 29 drivers were used to calculate the standardized value for driving workload with statistical significance. The method suggested in this study, which analyzes the driving workload by using brainwaves, can have various applications in HVI research and road design. In the future, the collected data on driver brainwaves will be used to standardize the driving workload based on the type of road, such as straight, right-turn, curve and tunnel driving, and driving patterns, such as acceleration and deceleration.

ACKNOWLEDGEMENT—This study was carried out as part of the industrial source technology development project under the Ministry of Knowledge Economy.

REFERENCES

- Andreassi, J. L. (1995). *Psychophysiology: Human Behavior and Physiological Response*. 3rd edn. Lawrence Erlbaum Associates.
- Kim, J. Y. and Yoon, S. Y. (2002). Comparison of methods to judge a driver's psychological and physiological states through biological signals. *Proc. Spring Conf. Ergonomics Society of Korea*, 309–312.
- Kim, J. Y., Park, J. S., Lee, H. Y. and Yoon, S. Y. (2004). Analysis of psychologically optimal driving state through measurement of physiological signals. *J. Korean Society for Emotion and Sensibility* **7**, 3, 27–35.
- Kim, J. Y., Yoon, S. Y., Park, J. S. and Lee, H. Y. (2001). Pattern classification and diagnosis of driver's bio-signal on highway. *Proc. Spring Conf. Ergonomics Society of Korea*, 234–239.
- Kim, M. H., Son, J. W., Lee, Y. T. and Shin, S. H. (2010). Development of vehicle environment for real-time driving behavior monitoring system. *J. Ergonomics Society of Korea* **29**, 1, 17–24.
- Kim, M. H. and Son, J. (2011). On-road assessment of in vehicle driving workload for older drivers: Design guidelines for intelligent vehicles. *Int. J. Automotive Technology* **12**, 2, 265–272.
- Koo, T. Y., Kim, B. Y., Ji, S. H., Bae, C. H., Park, J. H. and Suh, M. W. (2009). A study on workload of using telematics while driving. *Trans. Korean Society of Automotive Engineers* **17**, 2, 26–33.
- Naim, W. G., Mironer, Jr. M., Wang, J. S. and Knipling, R. R. (1995). Synthesis Report: Examination of Target Vehicular Crashes and Potential ITS Countermeasures. DOT HS 808 263. NHTSA U.S. Dept. Transportation.
- National Police Agency (1998). Statistice of Road Traffic Accidents in Japan. National Police Agency. 2475.
- Park, S. S., Hu, H. and Lee, W. S. (2010) A study on physiological signal changes due to distraction in simulated driving. *J. Ergonomics Society of Korea* **29**, 1, 55–59.
- Road Traffic Authority (ROTA) (2005). Road Traffic Accidents in Korea. Road Traffic Authority. 983.